Homework 1

Data Analysis and Machine Learning with Python

Problem description:

Please use two datasets and answer their corresponding questions. Each question requires an explanation of your thoughts and actions, along with relevant evidence or charts if possible. Additionally, not all questions have standard answers.

Dataset 1: housing_data.csv

Field Descriptions:

'Area': House area,

'No. of Rooms': Number of rooms (not including Bathrooms),

'No. of Bathrooms': Number of bathrooms,

'Location': Location of the house (Suburb, City Center, or Rural),

'Miles (dist. between NTU and house)': Distance between the house and the NTU,

'Rent Price per Month': Monthly rent price,

'Sell Price': House selling price

I. Part 1: Questions Related to Data Analysis

Q1. What steps will you take upon receiving this dataset before commencing data analysis?

Q2. If you inquire about Q1 from an AI tool, what responses will you receive? Do you find them reasonable? If not, how will you rectify it?

Q3. If you are restricted to renting a house, which one or ones will you select, and why?

Q4. Assuming you have enough funds to purchase a house, will you opt to continue renting or proceed with a purchase? If renting, which one will you choose? If buying, which one will you select? Why?

Q5. Are there any properties with rent or selling prices that seem unusually high or low? Why?

II. Part 2: Questions Related to the Use of AI Tools:

A. Did you use AI tools to assist in completing Q3 to Q5? (Please answer honestly)

- If you answered "No," please respond to questions A1 and A2.
- If you answered "Yes," please respond to questions B1 to B12 and provide your interaction history with the AI tool (you can share a webpage link or screenshots).

A1. Why did you choose not to use AI tools for assistance?

A2. If you were to use AI tools in the future, in what situations do you think you would use them?

B1. Which AI tool(s) did you use?

B2. With so many AI tools available, why did you choose this particular AI tool (or these AI tools) for assistance?

B3. What factors did you consider when selecting the AI tool? e.g., functionality, learning curve, accuracy, data security, etc.

B4. Did you compare the results of multiple AI tools? If so, what were the differences between the results, and what do you think caused these differences?

B5. How did you ensure that the output from the AI tool(s) was reliable?

B6. How did you design your prompt (command or question) to address Q3 to Q5? Did you adjust or iterate on this prompt?

B7. Was your prompt directly copied from the assignment question, or did you modify it based on the assignment requirements? What specific modifications did you make?

B8. When solving Q3 to Q5, how did you break down a large problem into smaller, manageable parts? Can you provide an example?

B9. How did you plan the steps to solve the Q3 to Q5? Was there a logical order to these steps?

B10. During the process of completing Q3 to Q5, did you encounter any challenges? How did you overcome them? How did the AI tool(s) help in this regard?

B11. After using the AI tool(s) to complete Q3 to Q5, did you learn any new data analysis techniques or knowledge?

B12. Throughout the process of solving Q3 to Q5, what aspects do you think AI tools cannot fully replace human thinking?

Dataset 2: family_data.csv

Field Descriptions:

'Family': Family ID,

'Member': Member (Adult or Child),

'Income': Annual income,

'Spend': Annual spending

I. Part 1: Questions Related to Data Analysis

Q1. Which family boasts the highest annual income, and which has the lowest? How do you ascertain this?

Q2. Which families do not possess adequate annual income to cover all members' spending? What is the maximum shortfall? How do you determine this?

Q3. Are there any single-parent families, where only one Adult is present? Are there any childless families? How do you discern this?

Q4. Do you suspect any errors within this dataset? Examples may include negative figures, missing or duplicate data, etc. Why?

II. Part 2: Questions Related to the Use of AI Tools:

A. Did you use AI tools to assist in completing Q1 to Q4? (Please answer honestly)

- If you answered "Yes," please respond to questions B1 to B12 and provide your interaction history with the AI tool (you can share a webpage link or screenshots).

B1. How did you translate complex problems into prompts that AI tools can understand?

B2. Before submitting your prompt, did you test some simple questions first to understand the tool's response style?

B3. Did you adopt a "step-by-step prompt" strategy? For example, did you first ask the AI to perform data cleaning before proceeding to analysis?

B4. Did your prompt include specific analysis objectives (e.g., generating charts, predicting trends, calculating specific metrics)?

B5. Before inputting data into the AI tool, did you perform any data cleaning? For instance, handling missing values or ensuring consistent formatting?

B6. Did you consider the source and quality of the data? For example, were there potential biases or noise in the data?

B7. Do you understand the logic behind the model chosen by the AI tool? For example, why is this model suitable for your data characteristics?

B8. When the AI tool provided multiple analysis results, how did you determine which result was more valuable?

B9. When the analysis results did not match your expectations, how did you adjust your analysis process?

B10. How did you avoid "blindly trusting AI tool results"?

B11. Did you summarize your analysis process, noting which methods were effective and which needed improvement?

B12. When facing a new problem, how do you decide whether to "calculate manually" or "rely on AI tools"?

Submission details:

please submit all the required files in a folder, named as "<studentID>_hw1", then zip it as a .zip file, named as "<studentID>_hw1.zip", kindly check the example below.

- required:
    - 1. your report, named as "<studentID>_report.pdf" (example: b12345678_report.pdf)
    - 2. your source code:
        - if you are using .ipynb, please submit as two files, named as "Housing_Main.ipynb" and "Family_Main.ipynb"
        - if you are using .py, please name your two main codes as "Housing_Main.py" and "Family_Main.py", all the other source code can be named as you pleased.
- optional:
    - 1. a README file, specifying how to execute your code, as well as your program's dependencies.
- not required:
    - 1. the datasets. Please assume the datasets are located under the same directory as your main code, **DO NOT** include the dataset in your submission. (20% score will be deducted on any violation)

example submission:

/b12345678_hw1.zip
---/b12345678_hw1
---/b12345678_report.pdf
---/Housing_Main.ipynb
---/Family_Main.ipynb
---/ ... <all other source code>
---/README.pdf (optional)

Homework policies:

- **deadline**: before class on 5th week
- We encourage you to discuss or ask questions at **Homework 1 Discussion**, but you can still email TAs and professor.
- You can discuss the homework with your classmates, but you **CANNOT** **plagiarize** your classmate's codes and reports. If we find **two files are exactly the same**, the **score** for these two files are **0**.
- The result in your report must be reproducible with your source code.